



Polyhedral Oligomeric Silsesquioxane/Multiwalled Carbon Nanotube (MWCNT/POSS) Hybrid Filler in Ethylene-Propylene-Diene Monomer/Acrylonitrile-Butadiene Rubber (EPDM/NBR) Nanocomposites: Preparation and Properties

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Abstract

In this study, ethylene-propylene-diene monomer/acrylonitrile-butadiene rubber (EPDM/NBR) nanocomposites were developed by incorporating a hybrid filler composed of polyhedral oligomeric silsesquioxane (POSS) and multiwalled carbon nanotubes (MWCNT) using a two-roll mill mixing technique. The research focused on evaluating cure characteristics, mechanical properties, abrasion resistance, compression set, swelling resistance, and crosslinking density of the prepared nanocomposites. The findings revealed that increasing the hybrid filler content led to a decrease in minimum torque, maximum torque, delta torque, and cure rate index, while scorch time and optimum cure time increased. Tensile strength and stress at 100% elongation improved with filler addition up to 4 phr but declined at higher concentrations. Hardness and tear strength generally increased with filler content, whereas elongation at break and rebound resilience showed a decreasing trend. Compression set values increased as filler content rose. Swelling resistance and abrasion resistance initially improved with the incorporation of 4 phr of hybrid filler but diminished at higher concentrations.

Keywords EPDM/NBR · POSS/MWCNT · Swelling resistance · Mechanical properties

1 Introduction

A key challenge in polymer vulcanizates is substituting traditional fillers like carbon black and silica, which are generally used in large quantities, with smaller amounts of nanofillers such as nanoclay, multiwalled carbon nanotubes (MWCNTs), and graphene to improve material properties [1–4]. This study focuses on polyhedral oligomeric silsesquioxanes (POSS) [5] and MWCNTs, two promising nanofillers. POSS, an organic-inorganic hybrid compound, is widely utilized in nanomaterial [6] and nanocomposite development [7, 8]. Its molecular structure, $R_n(\text{SiO}_{1.5})_n$, where R represents organic groups and n ranges from 8 to 12 depending on its configuration [9], forms a cage-like structure with excellent thermal and chemical stability [10]. POSS molecules inherently exist at the nanoscale (typically 3–5 nm) [11]; however, in bulk form, they tend to agglomerate and only disperse to the nanoscale when integrated into a solvent or polymer matrix. POSS-based nanocomposites have attracted considerable interest due to their versatile surface modification capabilities [12–14], which improve

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